

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - MEDIUM (Scenario Based Assessment)

COURSE CODE: CSA0904

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks: 50

ANNEXURE A

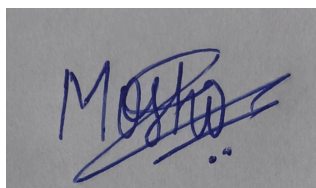
Scenario Based Assessment

Code of Conduct:

I Moshika M - 192424064 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding & Context Analysis	25	Thoroughly understands the scenario with accurate interpretation of all requirements	Clearly understands the scenario with minor gaps	Basic understanding; some aspects of the scenario unclear	Poor understanding or misinterpretation of the scenario
Application of Concepts	30	Applies appropriate concepts effectively to solve complex real-world problems	Applies relevant concepts correctly with minor errors	Applies basic concepts but with limited accuracy	Fails to apply appropriate concepts
Analytical & Decision-Making Skills	20	Demonstrates strong analysis and selects optimal solutions with justification	Good analysis with reasonable solutions	Limited analysis; solutions lack justification	Poor or no analysis; incorrect decisions

Solution Design & Approach	15	Well-structured, innovative, and feasible solution approach	Logical and structured solution with minor gaps	Basic solution with limited structure or feasibility	Unclear or incorrect solution approach
Clarity, Justification & Presentation	10	Clear, well-justified explanation with strong reasoning	Clear explanation with some justification	Limited clarity and weak justification	Poorly explained with no justification

ANNEXURE

Q1: Student Registration System

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

- Enumerate three Collection interfaces suitable for this system.
- Design a Collection structure (diagram/representation) to store students and their enrolled courses.
- Write a Java program using HashSet and HashMap with Generics to:
 - Add students
 - Map students to courses
 - Display data using Iterator

Q2: E-Commerce Cart System

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

- List three Collection classes applicable for cart management.
- Represent how cart items are stored using a List-based structure.
- Write a Java program using ArrayList and Iterator to:
 - Add items
 - Remove an item
 - Display all items

Q3: Library Management System

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

- Name three Map implementations in Java.
- Design a schema-like representation using Map for storing ISBN and book details.
- Write a Java program using HashMap to:
 - Insert book details
 - Retrieve a book
 - Display all books

Q4: Employee Data Processing (10 Marks)

Suppose that a company processes employee records and filters employees based on salary.

- List three advantages of Generics in Collections.
- Represent how employee data is stored using List with Generics.
- Write a Java program using Iterator to:
 - Store employee objects
 - Display employees with salary > 50,000

Q5: Online Voting System (10 Marks)

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

(a) Identify three suitable Collection types for this system.

(b) Design a structure using Set and Map for voters and vote counting.

(c) Write a Java program using HashSet and HashMap to:

- Record votes
- Prevent duplicate voting • Display results

Q1 Student Registration System

10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

- (a) Enumerate three Collection interfaces suitable for this system.
- (b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.
- (c) Write a Java program using HashSet and HashMap with Generics to:
 - Add students
 - Map students to courses
 - Display data using Iterator

ANSWER:

Collection interfaces:

1. Set → Ensures uniqueness of students. Since no two students can be duplicated in the system, a **Set** is ideal for storing student records.
2. List → Useful for maintaining ordered lists of courses. For example, if a student enrolls in multiple courses, the order of enrollment can be preserved.
3. Map → Provides key-value mapping. A student can be the key, and their enrolled courses can be the value, allowing efficient lookups.

Structure:

A **HashSet<Student>** ensures unique students. A **HashMap<Student, HashSet<Course>>** maps each student to their enrolled courses. This combination guarantees uniqueness and efficient retrieval.

CO2-AT1-Scenario Based Programs

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QUESTIONS

Q1 ✓
Student Registration System
10 marks

Q2 ✓
E-Commerce Cart System
10 marks

Q3 ✓
Library Management System
10 marks

Q4 ✓
Employee Data Processing
10 marks

Q5 ✓
Online Voting System
10 marks

5/5 answered

Q1 Student Registration System

10 marks

Suppose that a university system stores student data, where each student must be unique and can enroll in multiple courses.

- (a) Enumerate three Collection interfaces suitable for this system.
- (b) Design a Collection structure (diagram/representation) to store students and their enrolled courses.
- (c) Write a Java program using HashSet and HashMap with Generics to:
 - Add students
 - Map students to courses
 - Display data using Iterator

Your Code

```
1 import java.util.HashMap;
2 import java.util.HashSet;
3 import java.util.Iterator;
4 import java.util.Map;
5 import java.util.Set;
6 public class StudentRegistrationSystem {
7     public static void main(String[] args) {
8         Set<Student> students = new HashSet<>();
9         Map<Student, Set<String>> studentCourses = new HashMap<>();
10        Student s1 = new Student("S101", "Moshika");
```

Input

stdin input

Output


```
11      Student s2 = new Student("S102", "Arun");
12      Student s3 = new Student("S103", "Karthik");
13      students.add(s1);
14      students.add(s2);
15      students.add(s3);
16      studentCourses.put(s1, new HashSet<>());
17      studentCourses.put(s2, new HashSet<>());
18      studentCourses.put(s3, new HashSet<>());
19      studentCourses.get(s1).add("CO2");
20      studentCourses.get(s1).add("DSA");
21      studentCourses.get(s1).add("JAVA");
22      studentCourses.get(s2).add("CO2");
23      studentCourses.get(s2).add("DBMS");
24      studentCourses.get(s3).add("JAVA");
25      studentCourses.get(s3).add("OS");
26      Iterator<Student> studentIterator = students.iterator();
27      while (studentIterator.hasNext()) {
28          Student student = studentIterator.next();
29          System.out.println("Student: " + student);
30          Set<String> courses = studentCourses.get(student);
31          Iterator<String> courseIterator = courses.iterator();
32          System.out.print("Enrolled Courses: ");
33          while (courseIterator.hasNext()) {
34              System.out.print(courseIterator.next());
35              if (courseIterator.hasNext()) System.out.print(", ");
36          }
37          System.out.println("\n");
38      }
39  }
```

```
36         }
37         System.out.println("\n");
38     }
39 }
40 }
41 class Student {
42     private String id;
43     private String name;
44     public Student(String id, String name) {
45         this.id = id;
46         this.name = name;
47     }
48     public boolean equals(Object obj) {
49         if (this == obj) return true;
50         if (obj == null || getClass() != obj.getClass()) return false;
51         Student student = (Student) obj;
52         return id.equals(student.id);
53     }
54     public int hashCode() {
55         return id.hashCode();
56     }
57     public String toString() {
58         return id + " - " + name;
59     }
60 }
```

**Run****Save**

Output

Student: S102 - Arun

Enrolled Courses: CO2, DBMS

Student: S103 - Karthik

Enrolled Courses: JAVA, OS

Output

```
Enrolled Courses: JAVA, OS
```

```
Student: S101 - Moshika
```

```
Enrolled Courses: JAVA, DSA, CO2
```

Q2 E-Commerce Cart System

10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

- (a) List three Collection classes applicable for cart management.
- (b) Represent how cart items are stored using a List-based structure.
- (c) Write a Java program using ArrayList and Iterator to:
 - Add items
 - Remove an item
 - Display all items

ANSWER:

Collection classes:

1. ArrayList → Provides dynamic resizing and fast random access, making it suitable for cart operations.
2. LinkedList → Efficient for frequent insertions and deletions, useful when customers frequently update their cart.
3. Vector → Synchronized version of ArrayList, useful in multi-threaded environments where multiple users may access carts simultaneously.

Structure:

A `List<Product>` (commonly `ArrayList`) stores cart items in order. Each product can be added, removed, or displayed sequentially.

CO2-AT1-Scenario Based Programs

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QUESTIONS

Q1 ✓
Student Registration System
10 marks

Q2 ✓
E-Commerce Cart System
10 marks

Q3 ✓
Library Management System
10 marks

Q4 ✓
Employee Data Processing
10 marks

Q5 ✓
Online Voting System
10 marks

5/5 answered

Q2 E-Commerce Cart System

10 marks

Suppose that an online shopping system maintains a cart where products are added, removed, and displayed in order.

- (a) List three Collection classes applicable for cart management.
- (b) Represent how cart items are stored using a List-based structure.
- (c) Write a Java program using ArrayList and Iterator to:
 - Add items
 - Remove an item
 - Display all items

Your Code

```
1 import java.util.*;
2 public class EcommerceCart {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         ArrayList<String> cart = new ArrayList<>();
6         int n = sc.nextInt();
7         sc.nextLine();
8         for (int i = 0; i < n; i++) {
9             cart.add(sc.nextLine());
10        }
```

Input

```
3
Laptop
Mouse
Keyboard
Mouse
```

Output

```
Cart: [Laptop , Mouse, Keyboard]
Cart Items After removing an item:
Laptop
Keyboard
```




Q4

Employee Data Processing
10 marks



Q5

Online Voting System
10 marks



5/5 answered

Your Code

```
1 import java.util.*;
2 public class EcommerceCart {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         ArrayList<String> cart = new ArrayList<>();
6         int n = sc.nextInt();
7         sc.nextLine();
8         for (int i = 0; i < n; i++) {
9             cart.add(sc.nextLine());
10        }
11        System.out.println("Cart: " + cart);
12        String item = sc.nextLine();
13        for (int i = 0; i < cart.size(); i++) {
14            if (cart.get(i).equalsIgnoreCase(item)) {
15                cart.set(i, "");
16            }
17        }
18        System.out.println("Cart Items After removing an item:");
19        Iterator<String> it = cart.iterator();
20        while (it.hasNext()) {
21            String s = it.next();
22            if (!s.equals(""))
23                System.out.println(s);
24        }
25    }
26 }
```

Input

```
3
Laptop
Mouse
Keyboard
Mouse
```

Output

```
Cart: [Laptop , Mouse, Keyboard]
Cart Items After removing an item:
Laptop
Keyboard
```


Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

- (a) Name three Map implementations in Java.
- (b) Design a schema-like representation using Map for storing ISBN and book details.
- (c) Write a Java program using HashMap to:
 - Insert book details
 - Retrieve a book
 - Display all books

ANSWER:

Map implementations:

1. HashMap → Provides constant-time performance for insert and lookup, ideal for ISBN-based retrieval.
2. TreeMap → Maintains sorted order of keys, useful if books need to be displayed in ascending ISBN order.
3. LinkedHashMap → Maintains insertion order, useful for displaying books in the order they were added.

Schema:

`HashMap<String, Book>` where the key is ISBN (unique identifier) and the value is the book object containing details like title, author, and publisher.

CO2-AT1-Scenario Based Programs

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QUESTIONS

Q1 ✓
Student Registration System
10 marks

Q2 ✓
E-Commerce Cart System
10 marks

Q3 ✓
Library Management System
10 marks

Q4 ✓
Employee Data Processing
10 marks

Q5 ✓
Online Voting System
10 marks

5/5 answered

Q3 Library Management System

10 marks

Suppose that a library stores books with unique ISBN numbers and needs fast retrieval.

- (a) Name three Map implementations in Java.
- (b) Design a schema-like representation using Map for storing ISBN and book details.
- (c) Write a Java program using HashMap to:
 - Insert book details
 - Retrieve a book
 - Display all books

Your Code

```
1 import java.util.*;  
2 public class LibraryManagement {  
3     public static void main(String[] args) {  
4         Scanner sc = new Scanner(System.in);  
5         HashMap<Integer, String> library = new HashMap<>();  
6         int n = sc.nextInt();  
7         sc.nextLine();  
8         for (int i = 0; i < n; i++) {  
9             int isbn = sc.nextInt();  
10            sc.nextLine();
```

Input

```
3  
101  
Java  
102
```

Output

```
Retrieved Book: Python
```

```
Library Books:
```


Q4

Employee Data Processing

10 marks

**Q5**

Online Voting System

10 marks



5/5 answered

Your Code

```
1 import java.util.*;
2 public class LibraryManagement {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         HashMap<Integer, String> library = new HashMap<>();
6         int n = sc.nextInt();
7         sc.nextLine();
8         for (int i = 0; i < n; i++) {
9             int isbn = sc.nextInt();
10            sc.nextLine();
11            String book = sc.nextLine();
12            library.put(isbn, book);
13        }
14        int search = sc.nextInt();
15        if (library.containsKey(search))
16            System.out.println("Retrieved Book: " + library.get(search));
17        else
18            System.out.println("Book Not Found");
19        System.out.println("\nLibrary Books:");
20        for (Map.Entry<Integer, String> e : library.entrySet()) {
21            System.out.println(e.getKey() + " : " + e.getValue());
22        }
23    }
24 }
```

**Run****Save**

Input

Python

103

C++

102

Output

Retrieved Book: Python

Library Books:

101 : Java

102 : Python

103 : C++

Q4 Employee Data Processing

10 marks

Suppose that a company processes employee records and filters employees based on salary.

- (a) List three advantages of Generics in Collections.
- (b) Represent how employee data is stored using List with Generics.
- (c) Write a Java program using Iterator to:
 - Store employee objects
 - Display employees with salary > 50,000

ANSWER:

Advantages of Generics:

1. **Type Safety** → Prevents runtime errors by ensuring only specific types are stored in collections.
2. **No Casting Required** → Eliminates the need for explicit type casting when retrieving elements.
3. **Code Reusability** → Allows writing generic classes and methods that work with different data types.

Structure:

`List<Employee>` ensures that only employee objects are stored, avoiding accidental insertion of unrelated data.

CO2-AT1-Scenario Based Programs

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QUESTIONS

Q1 ✓
Student Registration System
10 marks

Q2 ✓
E-Commerce Cart System
10 marks

Q3 ✓
Library Management System
10 marks

Q4 ✓
Employee Data Processing
10 marks

Q5 ✓
Online Voting System
10 marks

5/5 answered

Q4 Employee Data Processing

10 marks

Suppose that a company processes employee records and filters employees based on salary.

- (a) List three advantages of Generics in Collections.
- (b) Represent how employee data is stored using List with Generics.
- (c) Write a Java program using Iterator to:
 - Store employee objects
 - Display employees with salary > 50,000

Your Code

```
1 import java.util.ArrayList;
2 import java.util.Iterator;
3 import java.util.List;
4 public class Main {
5     public static void main(String[] args) {
6         List<Employee> employees = new ArrayList<>();
7         employees.add(new Employee(101, "Asha", 45000));
8         employees.add(new Employee(102, "Karthik", 65000));
9         employees.add(new Employee(103, "Mohan", 52000));
10        employees.add(new Employee(104, "Priya", 30000));
11        Iterator<Employee> it = employees.iterator();
12        System.out.println("Employees with salary > 50,000:");
```

Input

stdin input

Output

```
Employees with salary > 50,000:
Employee{id=102, name='Karthik',
salary=65000.0}
Employee{id=103, name='Mohan',
salary=52000.0}
```



Online Voting System
10 marks

5/5 answered

```
6      List<Employee> employees = new ArrayList<>();
7      employees.add(new Employee(101, "Asha", 45000));
8      employees.add(new Employee(102, "Karthik", 65000));
9      employees.add(new Employee(103, "Mohan", 52000));
10     employees.add(new Employee(104, "Priya", 30000));
11     Iterator<Employee> it = employees.iterator();
12     System.out.println("Employees with salary > 50,000:");
13     while (it.hasNext()) {
14         Employee emp = it.next();
15         if (emp.salary > 50000) {
16             System.out.println(emp);
17         }
18     }
19 }
20 }
21 class Employee {
22     int id;
23     String name;
24     double salary;
25     Employee(int id, String name, double salary) {
26         this.id = id;
27         this.name = name;
28         this.salary = salary;
29     }
30     public String toString() {
31         return "Employee{id=" + id + ", name='" + name + "', salary=" +
32     }
33 }
```

Output

```
Employees with salary > 50,000:
Employee{id=102, name='Karthik',
salary=65000.0}
Employee{id=103, name='Mohan',
salary=52000.0}
```


Q5 Online Voting System

10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

- (a) Identify three suitable Collection types for this system.
- (b) Design a structure using Set and Map for voters and vote counting.
- (c) Write a Java program using HashSet and HashMap to:
 - Record votes
 - Prevent duplicate voting • Display results

ANSWER:

Collection types:

1. HashSet → Ensures unique voters, preventing duplicate voting.
2. HashMap → Stores candidates as keys and vote counts as values, allowing efficient tallying.
3. TreeMap → Provides sorted results, useful for displaying vote counts in order.

Structure:

`HashSet<Voter>` ensures each voter votes only once. `HashMap<Candidate, Integer>` stores vote counts.

CO2-AT1-Scenario Based Programs

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QUESTIONS

Q1 ✓
Student Registration System
10 marks

Q2 ✓
E-Commerce Cart System
10 marks

Q3 ✓
Library Management System
10 marks

Q4 ✓
Employee Data Processing
10 marks

Q5 ✓
Online Voting System
10 marks

5/5 answered

Q5 Online Voting System

10 marks

Suppose that an online voting system ensures one vote per user and counts votes efficiently.

- (a) Identify three suitable Collection types for this system.
- (b) Design a structure using Set and Map for voters and vote counting.
- (c) Write a Java program using HashSet and HashMap to:
 - Record votes
 - Prevent duplicate voting • Display results

Your Code

```
1 import java.util.*;
2 public class OnlineVotingSystem {
3     private static Set<String> votedVoters = new HashSet<>();
4     private static Map<String, Integer> voteCounts = new HashMap<>();
5     public static void main(String[] args) {
6         String[] candidates = {"Alice", "Bob", "Charlie"};
7         for (String c : candidates) {
8             voteCounts.put(c, 0);
9         }
10        castVote("V1001", "Alice");
11        castVote("V1002", "Bob");
12        castVote("V1003", "Alice");
```

Input

stdin input

Output

```
Vote recorded: V1001 -> Alice
Vote recorded: V1002 -> Bob
Vote recorded: V1003 -> Alice
Duplicate vote rejected for voter:
V1001
```




Online Voting System
10 marks

5/5 answered

```
5 public static void main(String[] args) {
6     String[] candidates = {"Alice", "Bob", "Charlie"};
7     for (String c : candidates) {
8         voteCounts.put(c, 0);
9     }
10    castVote("V1001", "Alice");
11    castVote("V1002", "Bob");
12    castVote("V1003", "Alice");
13    castVote("V1001", "Charlie");
14    displayResults();
15 }
16 private static void castVote(String voterId, String candidate) {
17     if (votedVoters.contains(voterId)) {
18         System.out.println("Duplicate vote rejected for voter: " +
19             voterId);
20         return;
21     }
22     votedVoters.add(voterId);
23     voteCounts.putIfAbsent(candidate, 0);
24     voteCounts.put(candidate, voteCounts.get(candidate) + 1);
25     System.out.println("Vote recorded: " + voterId + " -> " + candidate);
26 }
27 private static void displayResults() {
28     System.out.println("\n--- Voting Results ---");
29     for (Map.Entry<String, Integer> entry : voteCounts.entrySet())
30         System.out.println(entry.getKey() + " : " + entry.getValue());
31 }
32 }
```

Output

```
Vote recorded: V1001 -> Alice
Vote recorded: V1002 -> Bob
Vote recorded: V1003 -> Alice
Duplicate vote rejected for voter:
V1001
```